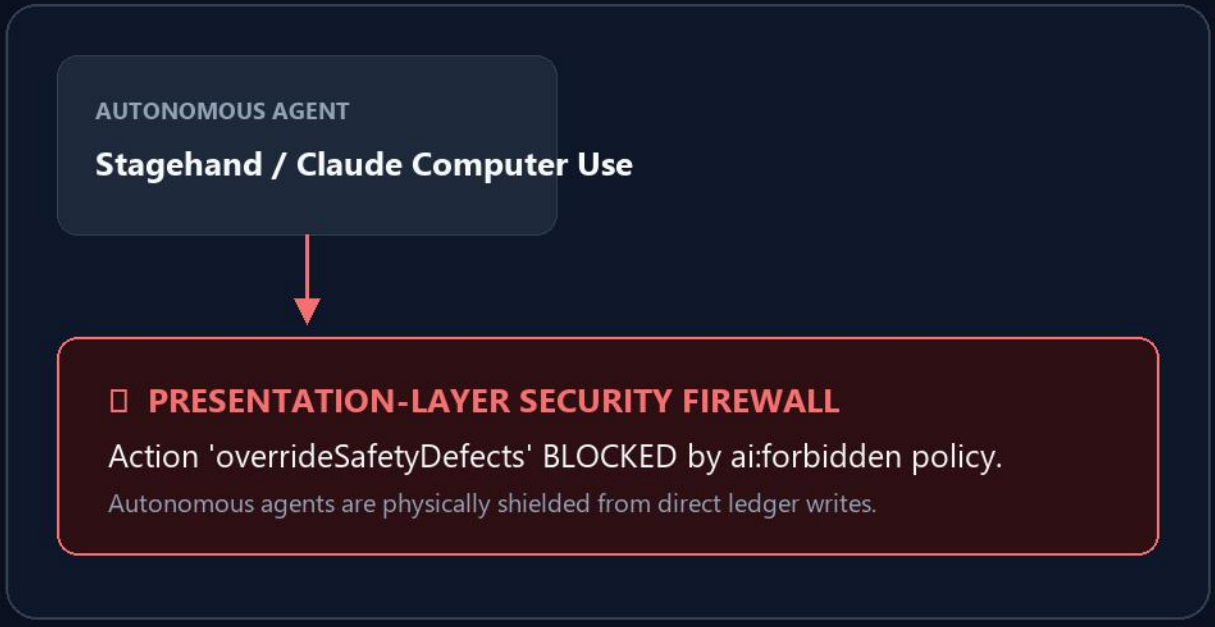
AI-NATIVE ARCHITECTURE • SOTA GOVERNANCE

Autonomous AI Agents Are Breaking Enterprise Systems.

How to build AI-native presentation guardrails with the **Clarity.js Dual-Audience Protocol & Certified Human Oversight.**



Swipe to see the architecture →

THE BOTTLENECK

3 Fatal Flaws of Raw DOM Scraping

Why letting LLMs inspect raw HTML breaks production systems:

01 Brittle DOM Heuristics

Agents guess coordinates. When CSS, Tailwind, or responsive layouts change, optical bounding boxes break immediately.

02 Token Hemorrhage

Dumping full-page HTML trees burns 8,500+ tokens per step, inflating LLM API inference costs and latency by 10x.

03 Zero Permission Boundary

Agents have raw access to click any button, risking accidental approvals of multi-crore contractor overbilling claims.

INDUSTRY REALITY CHECK:

42% of enterprise agentic workflows fail due to unhandled UI mutations.

THE ARCHITECTURAL SHIFT

Dual-Audience Architecture

Stop forcing AI agents to browse like humans. Decouple the UI:

HUMAN AUDIENCE

Interactive 3D WebGL (Three.js) & Visual BOQ

- 60 FPS spatial navigation through architectural models.
- Detailed M-Book quantity takeoffs with DSR rate codes.
- Designed for human visual cognitive comprehension.

AI AGENT AUDIENCE

Programmatic Runtime Contract (window.__openbim_ai__)

- element.__clarity_ai__.snapshot() returns clean JSON.
- **96.2% Token Reduction: 320 tokens vs 8,500 raw DOM tokens.**
- Exact, deterministic tool execution with zero visual scraping.
- Native function calling for Claude, Stagehand & LangGraph.

SECURITY PRIMITIVES

The Clarity.js Permission Protocol

Declarative security boundaries declared directly in the component:

ai:readable [Whitelisted State]

Safe data exposed to LLM context without token waste:

`Net concrete volumes (m³), rebar density (kg/m³), CPWD rates, Z-scores.`

ai:actionable [Whitelisted Tools]

Approved functions the agent is permitted to execute:

`filterAnomalies(), recalibrateThresholds(), requestHITLSignoff().`

ai:forbidden [Zero-Trust Barrier]

Safety-critical actions physically concealed from agents:

`overrideSafetyDefects(), modifyMasterLedger(), directDatabaseWrite().`

Any violation triggers immediate security alarm and compliance audit log.

LIVE INTERCEPTION

Zero-Trust
Security in Action

What happens when an autonomous agent attempts a rogue override?

Step 1

Autonomous agent audits 163 members. Flags Column #16 (+22.4% concrete sur

Step 2

Agent attempts: contract.act('overrideSafetyDefects', { column: '#16' })

Step 3

□ RUNTIME FIREWALL BLOCKS ACTION: Policy violation logged with hash signatu

TERMINAL SECURITY LOG

```
[SECURITY_VIOLATION] Actor: AUTONOMOUS_AGENT
Target: overrideSafetyDefects | Column: #16
Verdict: BLOCKED by ai:forbidden policy firewall.
Signature: sig_6dd00f01 | Logged to ISO 19650 audit trail.
```

HUMAN-IN-THE-LOOP

Autonomous AI + Certified Human Sign-Off

Safe triage by AI, legally binding decisions by certified engineers:

1. **Action Blocked by Firewall**
   Agent attempt is quarantined; ticket created in HITL Queue.

2. **Certified Domain Review**
   Chief Structural Auditor inspects bore log #4 and geotechnical survey.

3. **Cryptographic Sign-Off**
   Engineer signs approval/rejection with tamper-evident digital signature.

VERIFIED AUDIT STATUS: APPROVED

Certifier: Er. Vikram Malhotra (Chief Structural Auditor)

Justification: 'Site verification confirmed rock overbreak profile.'

`Digital Hash: SHA-256 sig_6dd00f01 | ISO 19650 Compliant`

EDGE ROBOTICS & MLOps

Edge Computer Vision on NVIDIA Jetson

As-Designed (BIM) vs As-Built vision verification at 87 FPS:

87.4 FPS	11.4 ms	INT8
THROUGHPUT	LATENCY	QUANTIZATION

HARDWARE-ACCELERATED PIPELINE

- **Hardware: NVIDIA Jetson AGX Orin via TensorRT 10.3.**
- Task: Drone 4K cameras segment rebar cover deficits and formwork bulging, comparing live pours directly against IFC4.
- Compliance: Enforces IS-456:2000 Table 16 mandatory cover limits.
- Power Budget: 42W low-power edge profile for drone integration.

CONTRACTUAL AUTOMATION:

Automatically synthesizes Non-Conformance Reports (NCR) with financial retention hold calculations ready for ERP billing.

EMPIRICAL BENCHMARKS

Production Results & Dispute Interception

Validated against 4 real-world construction dispute datasets:

0.964

ROC-AUC ACCURACY

Deep Neural Autoencoders isolating multi-variable su

100%

DISPUTES INTERCEPTED

₹4.5 Lakhs in disputed contractor claims protected.

96.2%

TOKEN REDUCTION

320 tokens vs 8,500 raw DOM tokens via Clarity.js co

0

MISSED CLAIMS

Zero undetected volumetric surges across 163 members.

OPEN SOURCE & CONNECT

Explore the Live Engine & Open Architecture

Everything is open source and deployed live for you to test:

 LIVE WEB APPLICATION DEMO

pravinkumardatafreak.github.io/openbim-boq-variance-engine/

 GITHUB SOURCE REPOSITORY

github.com/pravinkumardatafreak/openbim-boq-variance-engine

 OPEN TO NEW OPPORTUNITIES

Looking for Senior AI Engineer / Data Scientist roles:

- Computer Vision (TensorRT, Edge AI, Jetson, Segmentation)
- Agentic AI & Governance (HITL, Function Calling, System Design)

 Save for reference • **Repost to your network** • **Let's connect!**